Production Planning and Control

Calibrates Everything

Production Planning and Control (PPC)

Core Definition: The strategic engine that translates a high-tech prototype or code architecture into a scalable, commercially viable market product.

Traditional Industrial PPC

Primary Focus:

- Maximizing high-volume factory efficiency
- Reducing overall cost per unit
- Optimizing steady-state operational workflows

Technopreneurial PPC

Driven by Survival, Agility & Extreme Uncertainty

Coordinating limited critical startup resources:

- **Capital** & financial runway
- **Human Talent** & core expertise
- **Time** to deployment

Ultimate Goal: Deploy evolving tech without running out of cash before achieving **Product-Market Fit**.

Production Planning (The Vision and Strategy)

Before spending capital, the technopreneur acts as an architect to answer: ***What are we building, how will we build it, and when do we scale?***

Agile Demand Forecasting

01 / Market Signals

Instead of relying on historical sales data (which a tech startup does not have), planning relies on pre-orders, crowdfunding metrics, or pilot group signals.

Design for Manufacturing

02 / Scalability (DFM)

Mapping how software loops or electronic components scale up from single bench prototypes to thousands of units seamlessly.

Routing & Sourcing

03 / Resource Optimization

Deciding what components to make in-house versus what to contract out to overseas or specialized regional manufacturing vendors to keep initial asset costs low.

Production Control (The Execution and Adaptation)

Once the green light is given, control acts as the real-time radar to answer: ***How do we adapt to immediate disruptions while preserving our financial runway?***

Shop Floor & Micro-batch

01 / Process Agility

Pushing out minor engineering or software adjustments mid-production without stopping the entire line.

Dynamic Inventory Tracking

02 / Lean Stock

Keeping buffer stock minimal (Just-In-Time) because tech parts can become obsolete overnight or market demands can pivot drastically.

Bottleneck & Burn-Rate

03 / Risk Remediation

Real-time problem solving when custom microchips get delayed at customs or a server cloud capacity maxes out.

The Paradigm Shift

Element	Traditional Manufacturing PPC	Technopreneurship PPC
Primary Goal	**Minimize Unit Cost** via extreme replication	**Maximize Learning Rate** and speed-to-market
Tooling Setup	Heavy upfront tooling (e.g., steel molds)	Soft/Flexible tooling (e.g., 3D printing, CNC machining)
Material Risks	Storage space and handling overhead	**Component obsolescence** and sudden product pivots
Supply Chain	Rigid, long-term vendor contracts	Hybrid, fluid contract manufacturing ecosystem

Traditional Manufacturing PPC: Maximize Efficiency & Minimize Unit Cost

In a traditional setup, the market demand is predictable, and the product design is entirely frozen. The primary goal is **Economies of Scale**—making the exact same item millions of times as cheaply, quickly, and flawlessly as possible.

Metric of Success

01 / Cost & Efficiency

Unit Cost & OEE: Overall Equipment Effectiveness is paramount. If a machine sits idle, money is being lost.

Core Strategy

02 / High Upfront Investment

Diluting Fixed Costs: Schedules massive production runs to dilute fixed costs like expensive steel molds or automated assembly lines.

The Philosophy

03 / Zero Waste & Speed

"We know exactly what the customer wants. Our job is to build it with zero waste and maximum speed."

Technopreneurship PPC: Maximize Learning Rate & Speed-to-Market

In a tech startup, you are dealing with extreme uncertainty. You don't know if the market actually wants your product yet, and your design is constantly evolving. Therefore, the primary goal is **Agile Validation**—getting small batches to customers fast to learn what works before running out of money.

Metric of Success

01 / Speed & Survival

Time-to-Market & Cash Runway Preservation: Success is measured by how fast you can build, measure, and pivot based on customer data.

Core Strategy

02 / Micro-Batches

Low Upfront Investment: Schedules micro-batches (using 3D printing, CNC machining, or flexible cloud software architectures) so if a design fails, capital isn't wasted on inventory.

The Philosophy

03 / Rapid Testing

"We think we know what the customer wants, but we need to test it immediately. Our factory must be flexible enough to change tomorrow."

Strategic Comparison

Strategic Dimension	Traditional Manufacturing PPC	Technopreneurship PPC
Primary Driver	**Cost Reduction:** How do we make this 2% cheaper?	**Speed & Flexibility:** How do we get this to users next week?
The Nightmare Scenario	**A Machine Breakdown:** Stopping a high-speed assembly line costs thousands per minute.	**The Wrong Product:** Spending 6 months perfectly mass-producing an item nobody wants to buy.
Inventory View	**Asset/Buffer:** Raw materials are bought in bulk to secure steep volume discounts.	**Liability/Risk:** Materials are kept tight (Just-In-Time) because tech components become obsolete fast.
Capacity Focus	**High Utilization:** Machines and labor must be running at 90%+ capacity at all times.	**Excess/Fluid Capacity:** Keeping setups modular so production can instantly scale up, down, or pivot.

Core Pillars of Technopreneurial PPC

Pillar 01 / Dynamic BOM Management

Bill of Materials (BOM) Lifecycle Management

In technopreneurship, a product's bill of materials is alive. A change in an electronic sensor's availability can break the entire enclosure design.

Technopreneurial PPC leverages digital, integrated environments to instantly flag how engineering changes alter component costs and production timelines.

Pillar 02 / Modular Stage Scaling

Startups cannot afford automated assembly lines on day one. Technopreneurial PPC plans production in distinct stages matching fundraising milestones:

Stage 1: Alpha/Beta Stage

Single unit hand-assemblies for lab testing.

Stage 2: Pilot Run Stage

Short batches (10–100 units) utilizing temporary tooling to test marketplace feedback.

Stage 3: Mass Customization Stage

Offloading to a global contract partner once consumer traction justifies the investment.

Core Pillars of Technopreneurial PPC

Pillar 03 / Cash Runway Alignment

Cash Runway Alignment

"In technopreneurship, cash is the ultimate constraint."

Traditional PPC optimization targets machine uptime. However, hardware startups must subordinate machine efficiency to liquidity.

Production schedules must strictly align with funding disbursements to preserve operational runway.

Operational vs. Financial Trade-off

The Capital Freeze Risk

If the master schedule delivers parts 3 months prior to assembly, it prematurely locks up critical working capital.

Impact: Deprives marketing and R&D teams of vital funds required to keep the startup operational.

The Technopreneurial PPC Solution

Material drops and component orders are synchronized precisely with investor funding tranches and capital cycles.

Result: Maximize cash runway length without compromising production assembly timelines.

The Startup Sourcing Mix (Make vs. Buy vs. Hack)

Startups rarely build their own factories on day one. Learn how to orchestrate a fragmented, global network of suppliers.

01 / CONTRACT MANUFACTURING

CM Selection

"Tier 1 vs. Tier 3 CMs"

Evaluating and negotiating with appropriate contract manufacturing partners based on production scale and capital constraints.

02 / HYBRID FRAMEWORK

BYOP Framework

"Bring Your Own Parts"

Managing procurement risks for long-lead-time or critical custom silicon components independently and shipping directly to assembly partners.

03 / RAPID PROTOTYPING

The MVP Ecosystem

"Rapid Fabrication Services"

Utilizing rapid-turnaround fabrication & assembly services—such as PCBWay for electronics or Protolabs for injection molding—to handle short-run productions.

The Technopreneur's Dilemma: The Transition Path

For a technopreneur, starting directly with a Tier 1 manufacturer is almost always a mistake (and usually impossible due to capital constraints). However, relying entirely on a Tier 3 shop long-term can limit your ability to scale when your product experiences explosive market adoption.

01 / INITIAL MVP & PILOT RUNS

Launch at Tier 3

"Low Risk & Engineering Collaboration"

Use a Tier 3 manufacturer to build your initial Minimum Viable Product (MVP) and pilot runs. The proximity, lower financial risk, and engineering collaboration allow you to iron out design defects without burning through cash.

02 / HIGH-VOLUME PRODUCTION

Scale to Tier 2/1

"Verified Fit & Cost Reduction"

Once your product achieves verified product-market fit, stable engineering designs, and predictable high-volume demand, leverage your manufacturing data to transition to a larger tier partner to slash your unit production costs.

Supply Chain Financial Engineering & Capital Preservation

In technopreneurship, PPC decisions directly dictate cash runway. Learn to sync material drops with venture capital or crowdfunding cash flows.

01 / C2C CYCLE

Cash-to-Cash (C2C)

"Cycle Optimization"

Calculating exactly how many days capital is trapped in raw materials before it returns as revenue.

02 / MOQ STRATEGY

Minimum Order Quantity

"Negotiation Strategies"

Balancing the penalty of higher per-unit costs against the existential risk of tying up survival capital in mass inventory.

03 / OBSOLESCENCE

Component Obsolescence

"Modular BOM Design"

Designing a Bill of Materials (BOM) with modular, alternative parts so that if a specific chip goes out of stock, the entire production planning cycle does not freeze.

Cash-to-Cash (C2C) Cycle — also known as the **Cash Conversion Cycle (CCC)**

A financial metric that measures the exact number of days it takes for a startup to convert its cash outflows (invested in raw materials or manufacturing labor) back into cash inflows (revenue from customer sales).

Calculated using three operational metrics: **C2C Cycle = DIO + DSO - DPO**

+ DIO | INVENTORY

Days Inventory Outstanding (DIO)

+ ADDED TO CYCLE

The average number of days raw parts, work-in-progress units, and finished goods sit in your warehouse or factory floor before being sold.

+ DSO | RECEIVABLES

Days Sales Outstanding (DSO)

+ ADDED TO CYCLE

The average number of days it takes to collect cash from customers after a sale is made (e.g., immediate cash on a web store vs. 30-day payment terms from a retail store).

- DPO | PAYABLES

Days Payable Outstanding (DPO)

- SUBTRACTED FROM CYCLE

The average number of days your startup takes to actually pay its suppliers and contract manufacturers after receiving an invoice.

The Technopreneur's Strategy vs. Traditional PPC

Sourcing Strategy	Traditional PPC (Cost Focus)	Technopreneurship PPC (Cash Focus)
Procurement Strategy	Buys massive bulk orders to get a 10% unit discount.	Buys smaller micro-batches to avoid locking up cash.
Inventory State (DIO)	**High.** Thousands of units sit in storage for months.	**Low.** Just-In-Time parts arrive right before assembly.
Supplier Terms (DPO)	Standard corporate terms.	Negotiates deferred payments or structural staging.
C2C Cycle Impact	**Long C2C.** Cash is trapped for 90–120+ days.	**Short/Negative C2C.** Cash is freed up immediately.

Quality Control (QC) in Production Planning and Control (PPC)

TRADITIONAL APPROACH

Traditional QC Model

STATIC & END-OF-LINE

Relies on "frozen" product designs, strict mass-production tolerances, and statistical sampling at the end of a long assembly line.

Assumes fixed engineering specs and uniform batch manufacturing over high volumes.

THE STARTUP RISK

The Hardware Danger

HIGH-RISK BOTTLENECK

Waiting until the end of production creates a massive point of failure.

A single software bug, uncalibrated sensor, or minor defect can ruin an entire batch.

Risks instantly bankrupting the company through mass recalls or scrapped inventory.

MODERN SOLUTION

Technopreneurial QC

AGILE & CONTINUOUS

Agile, continuous, and digitally integrated quality management system.

An active software & hardware feedback loop catching flaws early.

Adapts seamlessly to shifting engineering designs and preserves startup runway.

Hardware-in-the-Loop (HIL) Assembly Testing

WHY INSPECTION FAILS

The Hybrid Challenge

PHYSICAL + FIRMWARE

Modern hardware devices are almost always hybrid products combining physical components with embedded software or firmware.

A circuit board might look visually flawless under a microscope, but the microcontroller could still fail to boot.

THE TACTIC

Assembly-Line Jigs

AUTOMATED FIXTURES

Technopreneurs design custom, automated test fixtures (often called "test beds" or "jigs").

Placed directly onto the assembly line for continuous, real-time quality assurance.

HOW IT WORKS

Instant Diagnostics

INLINE VERIFICATION

1. Clamped: Assembled board is secured into jig immediately.

2. Flashed & Tested: Jig flashes firmware and runs circuit diagnostics.

3. On-Spot Rejection: Failed parts are rejected before permanent housing installation.

Gateways for Evolving Designs (Dynamic ECOs)

THE CHALLENGE

Evolving Designs

MID-PRODUCTION ECOs

In tech startups, product designs change constantly based on early customer feedback (e.g., swapping Bluetooth chips).

Traditional QC halts the entire factory line to rewrite compliance rules.

THE TACTIC

Cloud PLM Tools

DYNAMIC LIFECYCLE MGT

Technopreneurial QC leverages cloud-based Product Lifecycle Management (PLM) platforms such as OpenBOM or Propel.

Enables seamless, centralized management of ECOs across engineering and manufacturing.

HOW IT WORKS

Digital Thread

REAL-TIME SYNC

1. Instant Push: Office changes push updated tolerances, schematics, and test specs.

2. Floor Sync: Factory screens and test jigs update automatically.

3. Zero Halts: Quality gates update dynamically without stopping production.

Post-Purchase Telemetry (The Continuous QC Loop)

THE SCOPE

Beyond the Dock

CONTINUOUS QUALITY

Technopreneurial QC doesn't stop when the product leaves the factory dock.

Quality control extends directly into real-world customer usage.

THE TACTIC

IoT Cloud Diagnostics

BUILT-IN TELEMETRY

Built-in IoT cloud diagnostics automatically collect device performance metrics.

Continuous telemetry bridges field performance directly back to engineering.

HOW IT WORKS

Closed-Loop QC

REAL-TIME FEEDBACK

1. Data Transmit: Device sends telemetry to engineering cloud when turned on.

2. Issue Detect: Battery-drain or sensor calibration errors are caught in real-world use.

3. PPC Update: Data immediately modifies assembly/calibration for next batch.

Traditional vs. Technopreneurial QC

Quality Dimension	Traditional Manufacturing QC	Technopreneurial PPC QC
Design Status	Fully frozen; tolerances rarely change	Fluid and iterative; design evolves via ECOs
Testing Timing	Post-assembly or end-of-line sampling	**In-line and automated** at every critical assembly step
Focus Area	Mechanical dimensions and cosmetic defects	**Functional validation** (Firmware integration, signal integrity)
Handling of Defects	Discarded or manually reworked at a high cost	**Rapid root-cause analysis**; immediate software/line adjustment

Executing QC: Local Technopreneur Strategies

For a Filipino technopreneur working with local **Tier 3 contract manufacturers** or utilizing government facilities like **DOST-AMCen**, implementing agile QC involves specific local strategies:

PROTOTYPING & TESTING JIGS

Build the Testing Jig Locally

OPEN-SOURCE & FABLABS

Startups frequently use open-source hardware (like **Arduinos** or **Raspberry Pis**) paired with local 3D-printed enclosures made at a local **DTI FabLab** to create custom testing jigs cheaply.

Deployment Strategy:

They hand this custom jig directly to their local manufacturer in **Laguna** or **Cavite** to fully automate the testing process on the production line.

REGULATORY COMPLIANCE

Navigate NTC & Compliance Early

NTC CERTIFICATION & DFM

In the Philippines, any device utilizing Wi-Fi, Bluetooth, or cellular networks must be certified by the **National Telecommunications Commission (NTC)**.

Integration Strategy:

Technopreneurial QC factors certification directly into the timeline by using pre-certified RF modules (like **ESP32**) during the Design-For-Manufacturing (**DFM**) phase to avoid failing regulatory quality checks later.

Thank you!